Progress Check

Quadratic equation	Roots of quadratic equation α and β	co-efficients of x^2, x and constants	Sum of Roots $\alpha+\beta$	Product of roots $\alpha\beta$	$-\frac{b}{a}$	$\frac{c}{a}$	Conclusion
$4x^2-9x+2=0$							
$\left(x-\frac{4}{5}\right)^2=0$							
$2x^2-15x-27=0$							

Example 3.43 If the difference between the roots of the equation $x^2-13x+k=0$ is 17 find k.

Solution $x^2-13x+k=0$ here, $a=1,\ b=-13,\ c=k$

Let α, β be the roots of the equation. Then

$\alpha+\beta=\frac{-b}{a}=\frac{-(-13)}{1}=13$ ...(1) Also $\alpha-\beta=17$ ...(2)

(1)+(2) we get, $2\alpha=30 \Rightarrow \alpha=15$

Therefore, $15+\beta=13$ (from (1)) $\Rightarrow \beta=-2$

But, $\alpha\beta=\frac{c}{a}=\frac{k}{1} \Rightarrow 15\times(-2)=k$ we get, $k=-30$

Thinking Corner

If the constant term of $ax^2+bx+c=0$ is zero, then the sum and product of roots are _____ and _____.

Example 3.44 If α and β are the roots of $x^2+7x+10=0$ find the values of

(i) $(\alpha-\beta)$ (ii) $\alpha^2+\beta^2$ (iii) $\alpha^3-\beta^3$ (iv) $\alpha^4+\beta^4$ (v) $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}$ (vi) $\frac{\alpha^2}{\beta}+\frac{\beta^2}{\alpha}$

Solution $x^2+7x+10=0$ here, $a=1,\ b=7,\ c=10$

If α and β are roots of the equation then,

$\alpha+\beta=\frac{-b}{a}=\frac{-7}{1}=-7;\ \alpha\beta=\frac{c}{a}=\frac{10}{1}=10$

(i) $\alpha-\beta=\sqrt{(\alpha+\beta)^2-4\alpha\beta}=\sqrt{(-7)^2-4\times10}=\sqrt{9}=3$

(ii) $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=(-7)^2-2\times10=29$

(iii) $\alpha^3-\beta^3=(\alpha-\beta)^3+3\alpha\beta(\alpha-\beta)=(3)^3+3(10)(3)=117$

(iv) $\alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2\alpha^2\beta^2=29^2-2\times(10)^2=641$ (from (ii), $\alpha^2+\beta^2=29$)

(v) $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{\alpha^2+\beta^2}{\alpha\beta}=\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}=\frac{49-20}{10}=\frac{29}{10}$

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